

Mazharul Islam Leon

Senior AI Engineer · Generative AI & Computer Vision Architect

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Professional Summary

Senior AI Engineer and solutions architect with **5+ years** designing and scaling production-grade **Computer Vision** and **Generative AI** systems serving **30M+ users** at national scale. Proven end-to-end ownership: real-time biometric verification at **50+ TPS** with 99.2% accuracy, **sub-50ms** edge inference, and autonomous **LLM agents** handling **10K+ daily queries**. Deep expertise in **agentic AI architecture** (LangChain, LangGraph, RAG, multi-agent systems), **MLOps**, and **edge AI**, with a record of converting PoCs into revenue-generating products.

Technical Skills

Languages	Python (Expert), C++, Java, SQL, Bash
Generative AI & Agents	LangChain, LangGraph, Google ADK, Agentic RAG, Multi-Agent Orchestration, Prompt Engineering, Guardrails, LLM Fine-tuning (LoRA, QLoRA, PEFT), Hugging Face
LLM Evaluation	RAGAS (Faithfulness, Answer Relevancy, Context Precision), LLM-as-a-Judge, BLEU, ROUGE
Deep Learning & CV	PyTorch, TensorFlow, CNNs, Vision Transformers (ViT), Object Detection (YOLO), Image Segmentation, OCR, Face Recognition, OpenCV
AI System Architecture	ML System Design, Microservices, Event-Driven Architecture, RESTful APIs, Real-Time Inference, High-Throughput & Low-Latency Systems
MLOps & Cloud	AWS (SageMaker), Docker, CI/CD (GitHub Actions), MLflow, Airflow, Model Deployment, Monitoring & Observability
Data & Storage	PostgreSQL, MySQL, Vector Databases (Chroma, Qdrant), ETL / Data Pipelines
Edge AI	NVIDIA Jetson, DeepStream, TensorRT, Model Optimization & Quantization

Professional Experience

Agripulse.ai

Oct 2025 – Jun 2026

Senior AI Engineer

Contract

- Designed and deployed an **agentic AI advisory chatbot** for farmers in Bangladesh on a LangChain orchestration core, delivering real-time agricultural guidance in Bengali across **10K+ daily queries**.
- Architected a **multi-agent system** with specialized sub-agents for crop disease diagnosis, weather forecasting, market price analysis, and fertilizer recommendation, enabling modular, extensible service composition.
- Implemented **agentic RAG pipelines** fusing structured agricultural databases with unstructured knowledge bases (FAO guidelines, local research) for context-aware, citation-grounded responses.
- Built an automated **LLM evaluation pipeline** using RAGAS metrics and LLM-as-a-Judge for continuous quality assurance, achieving **94% response accuracy**.
- Hardened prompts with few-shot examples, dynamic context injection, and **guardrails** to mitigate hallucinations in low-resource language settings.

RedDot Digital Ltd. – a subsidiary of Robi Axiata Limited

Aug 2021 – Jan 2026

Senior Software Engineer, IoT & Machine Learning

Full-time

- Architected a national-scale **AI biometric verification platform** (face recognition + camera-based fingerprint matching) serving **30M+ telecom subscribers**, sustaining **50+ TPS on CPU-only infrastructure** with **99.2% verification accuracy**.
- Designed and deployed an **OCR + face-matching eKYC platform** for **20M+ fintech users**, cutting customer onboarding from **48 hours to under 5 minutes** while maintaining full regulatory compliance.
- Built a real-time **video anomaly detection** system (hybrid YOLO + ML) on NVIDIA Jetson edge devices with DeepStream and TensorRT, achieving **sub-50ms inference at 30 FPS** in production environments.
- Engineered an end-to-end **computer vision pipeline** for automated tobacco leaf grading (classification + segmentation with MLflow experiment tracking), improving grading accuracy by **25%** and reducing manual processing time **3×**.
- Developed and optimized **LLM-powered chatbot systems** using LangChain, Google ADK, and vector databases (Chroma, Qdrant), autonomously handling **10K+ daily queries** and reducing support response time by **60%**.
- Spearheaded **5+ PoC initiatives**, owning solution architecture from requirements to production rollout and converting **3 into revenue-generating products**.

- Built data extraction and **ETL pipelines** from large-scale SQL dumps powering analytics and ML use cases.
- Performed data preprocessing and exploratory analysis supporting digital-health research and business insights.
- Collaborated with cross-functional teams to translate analytical findings into actionable product decisions.

Selected Projects

AI-Powered Biometric Verification System — Robi Axiata PLC

End-to-end biometric verification with face recognition and camera-based fingerprint matching for nationwide SIM registration — engineered for **30M users** at **50+ TPS**.

RedDot eKYC Platform — Telecash, Southeast Bank PLC

AI-powered eKYC for mobile banking featuring face identification, face matching, and OCR-based NID extraction — built to onboard **20M users** with regulatory-grade accuracy.

Publications

1. **M. I. Leon**, M. I. Iqbal, S. M. Azim, K. A. Mamun. "Predicting COVID-19 Infections and Deaths in Bangladesh using Machine Learning Algorithms." *ICICT4SD 2021*, IEEE Xplore.
2. **M. I. Leon**, M. I. Iqbal, S. Meem, F. Alahi, M. Ahmed, M. S. H. Mukta. "Dengue Outbreak Prediction from Weather Spatiotemporal Data using Deep Learning." *ICBDB 2021*, Springer.
3. S. Ahmed, **M. I. Leon**, S. Pal, M. R. H. Mondal. "Hybrid CNN-LSTM Transfer Learning for Dengue Diagnosis from Raman Spectroscopy Images." *ICTP 2023*, IEEE Xplore.
4. M. I. Iqbal, **M. I. Leon**, J. I. Rahat, N. H. Tonmoy, A. Ghosh. "Deep Learning Based Smart Parking Management System for Metropolitan City." *TENSYMP 2021*, IEEE Xplore.
5. K. A. Mamun et al. (incl. **M. I. Leon**). "Implementation of a Digital Healthcare Service Model for Preventive and Primary Healthcare in Rural Bangladesh." *IC4IR 2021*, Springer.
6. A. Islam et al. (incl. **M. I. Leon**). "EduBot: An Educational Robot for Underprivileged Children." *ICACTM 2019*, IEEE Xplore.

Education

United International University, Dhaka

Oct 2015 – Oct 2020

B.Sc. in Computer Science and Engineering — merit-based scholarships across multiple trimesters

Awards & Leadership

- **Star Developer Award** (Q1-2022 & Q3-2023), RedDot Digital Ltd. — top engineering recognition.
- **Runner-up**, SDG Hackathon 2.0, organized by Banglalink.
- Instructor, "Build with AI" — Google Developer Group (GDG) at UIU; Instructor, "Intro to Computer Vision" — Quantum AI.
- Vice President, UIU App Forum (Jan 2020 – Sep 2020); Member, Competitive Programming Club, UIU.